

The ULTIMATE Machine Knit Socks

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Why are these socks so special?

- **Knit Fast**

Machine knitters rejoice!

- **Great Fit**

Gusseted heel is constructed like a hand knit sock

- **Better Wearing**

The heel method eliminates the stress points in ordinary short-row machine knit heels.

- **Greater Durability**

The slip stitch heel (optional) wears longer.

- **Comfy Cushioned Foot Bed**

- **Mostly Seamless** *Sew-as-you-go method gives a smooth, comfortable join on a flat bed machine.*

- **Creative Possibilities**

The instep can be any stitch pattern

- **3 Versions** *Standard gauge, mid-gauge and bulky socks.*

- **Easy Customization** *to accommodate different foot sizes.*

- **Double Bed, Circular Version** *Also included (no seam in the foot) for the adventurous!*

- **Options** *Suggestions for ribbing and cuff.*



From Lucy:

"I first developed this pattern in the early 1990's because the traditional machine knitted sock really doesn't work with my high insteps. My hand knitted socks were a perfect fit but the knitting time wasn't always available. So I set myself to the task of duplicating the heel construction of the traditional hand knitted flap, turn and gusset arrangement. I still always have some hand knitted socks in progress as carry along work.

Sometimes I knit the foot part on the machine and do the ribbing as carry along/waiting around activity.

The first version of my machine knitted socks was a rather complicated double bed version, requiring lots of rehangng and reversing directions. It made a lovely sock, but I felt it also took me too long to knit. As time went on I realized what I really wanted, as well as the better heel construction, was a way to include a cushioned foot for sport socks. This led to the single bed version and I have been "tweaking" the pattern ever since.

I could never think of a name that adequately described them, hence the SWAN name. Sock Without A Name seemed appropriate. I hope you enjoy knitting these patterns and wearing your socks.

They will take you a bit longer to knit than the traditional machine knitted sock. They are actually simple to knit, though, once you get the hang of it. If you like the way they fit and feel as much as I do the little extra time is a good trade off. It's a LOT faster than doing it on 4 or 5 needles."



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Knit it Now purchased the rights to Lucy's book from her estate. This book contains her amazing construction techniques. It has been reformatted and updated with new photography, links to tutorials, updated diagrams and links to navigate within the document.

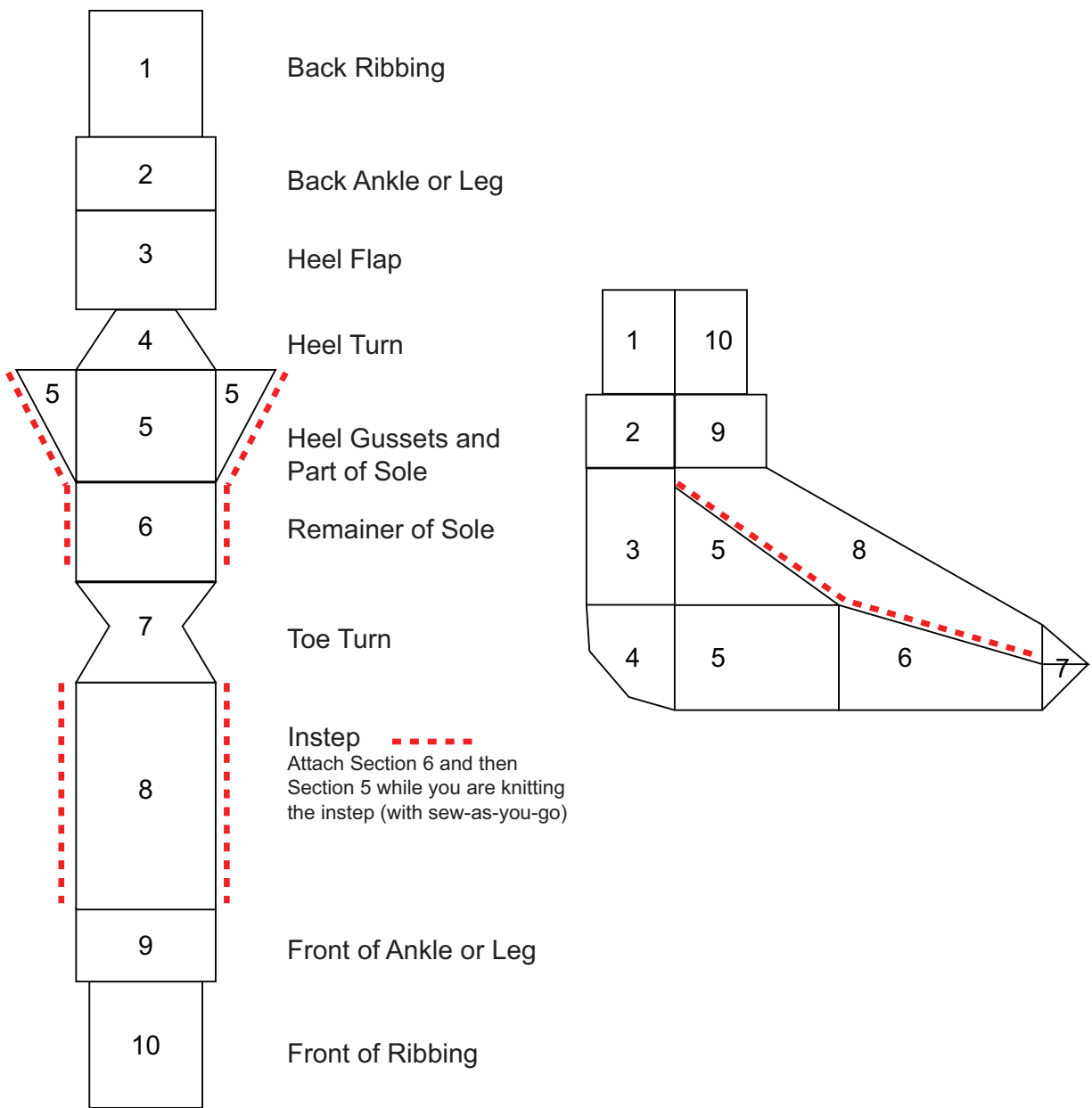
Ultimate Machine Knit Socks

Construction Method

Think of the sock as made up of several sections. These are numbered in the order in which you knit them.

Customize your socks by adding color and stitch patterns to the various sections.

If you do not like a seam on both sides of the ribbing, begin with waste yarn and a row of ravel cord. Knit section 2 through 9, then go back and knit the ribbing by hand or machine.



The Ultimate Machine Knit Socks

Materials and Skills Required

Machine:

Knit these socks with a standard, mid-gauge or bulky machine, preferably with ribber attachment. (Ribbing can be done by hand). Patterning capability—electronic or punchcard—is desirable, but not required as all patterning can easily be hand manipulated.

Tools

- 3-prong transfer tool
- waste yarn weight close to sock yarn and in a strongly contrasting color
- ravel cord (should come with the machine; use a strong, slippery cord such as bedspread cotton if it does not)
- weights
- small single bed cast-on comb (optional)

Yarn

Any yarn compatible with your machine. This book includes instructions for various machine gauges.

Skills

Basic machine knitting skills as described in your manual. These socks are not difficult but do require a number of basic skills. They are not a beginner project.

You need to be able to cast on, cast off, and decrease with a transfer tool.

If knitting the double bed version, you need to be able to knit tubular and set your machine for “U” knitting. You must be familiar with double bed shaping and transferring stitches between the beds.

You must be able to knit and measure tension swatches ACCURATELY.

Techniques used in the patterns:

- double E-wrap (Secured e-wrap)
- increasing short rows for heels
- short row and reverse short row for toes
- sew-as-you-go seaming method



Sue's Tip

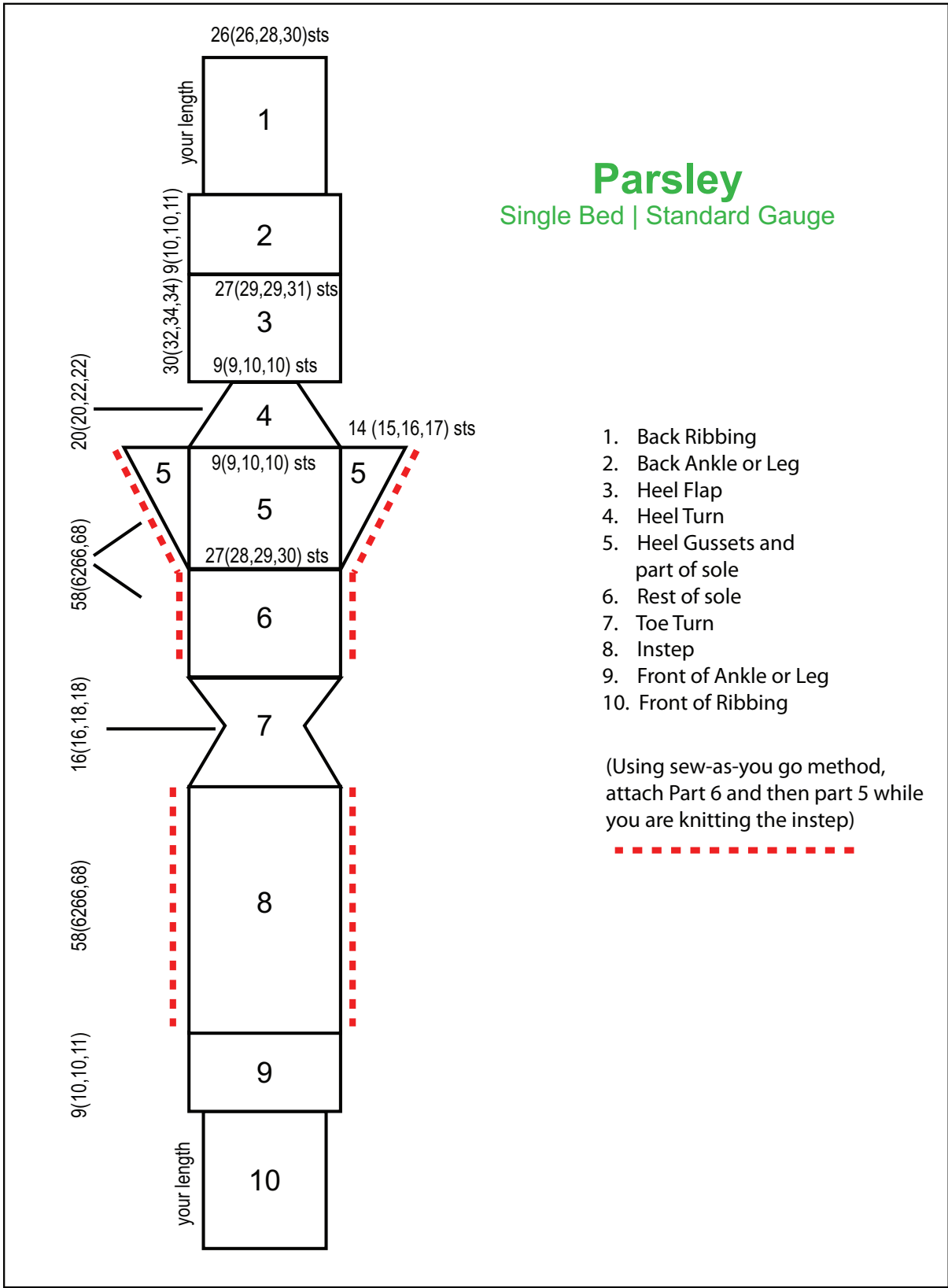
It's highly recommended you practice these skills BEFORE starting a sock.



Sue's Tip

Because this is a “design-it-yourself” pattern, options are offered in the construction method. As you become comfortable with knitting the sock you may want to try some of the options— or add your own!





The Ultimate Machine Knit Socks

Parsley Single Bed | Standard-Gauge

PARSLEY

WOMAN'S MEDIUM SOCK

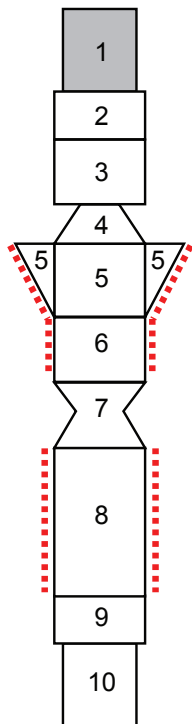
STANDARD GAUGE with cushioned foot, 2 color heel flap and slip stitch heel turn.

GAUGES:

7 (7½, 8, 8½) stitches per inch

9 (9½, 10, 10½) rows per inch

The measurements for this medium size are in **GREEN**. See instructions for [customizing your sock](#).



Section 1: Ribbing

EE wrap cast on double bed for 1x1 ribbing (or use LOOSE cast on method of your choice). Hang comb and weights. Knit desired length of ribbing.

(This is half of the ribbing. Section 10 is the other half. If you do not like a seam on each side of the ribbing, begin with waste yarn and a row of ravel cord and knit sections 2 through 9. Then go back and knit the ribbing by hand or machine.)

Cast on **(3.6" of stitches)**

Stitch Gauge	7	7½	8	8½	Your numbers
Cast on (sts)	26	26	28	30	



Sue's Tip

When knitting your first sock, we HIGHLY recommend working it entirely in stockinette. This will allow you to get a feel for the construction techniques without having to worry about stitch patterning.

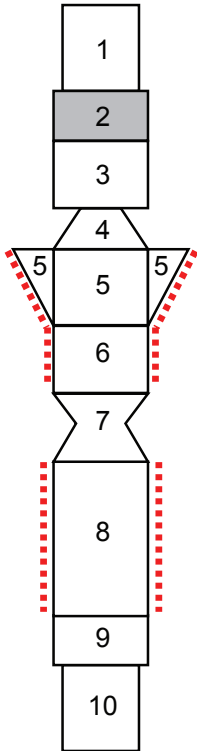


The Ultimate Machine Knit Socks

Parsley Single Bed | Standard-Gauge

Section 2: Ankle

Work in Stockinette



Row gauge	# Rows
9	9
9 ½	10
10	10
10 ½	11

Section 3: Heel Flap

- Stitch count must be odd; increase one stitch if necessary.

Stitch Gauge	7	7 ½	8	8 ½	
Stitch count	27	29	29	31	

- Program machine for heel stitch.
- Changing colors every 2 rows**, work for 2 ¼".

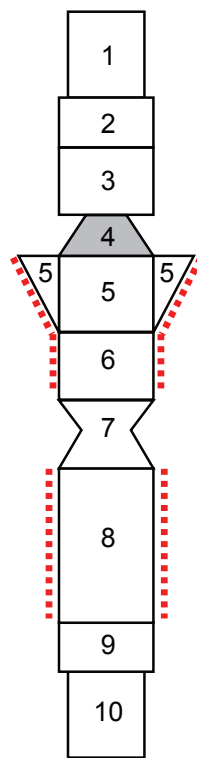
Row gauge	# Rows
9	30
9 ½	32
10	34
10 ½	34

Section 4: Heel Turn

In Stockinette, work INCREASING short rows with automatic wrap for 1 ½".

SEE TECHNIQUE following.

NOTE: Be sure to work an even number of rows so your heel turn will be the same on both sides.



Row Gauge	# Rows	Starting hold sts
9	20	9
9 ½	20	9
10	22	10
10 ½	22	10



The Ultimate Machine Knit Socks

Parsley Single Bed | Standard-Gauge

Section 5: Gussets And Start of Sole

Reset row counter to 000. Carriage on the right.

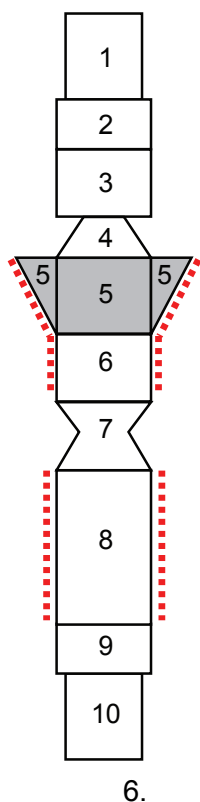
You will be knitting the foot length. The foot length for this size is (6.5" of rows)

1. On the carriage side (right), EE wrap cast on (2" of stitches)
2. Hang comb or weights on those stitches and knit 1 row.
3. Make sure all stitches knit.

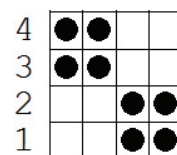
Stitch Gauge	7	7 ½	8	8 ½	
Add Stitches RIGHT →	14	15	16	17	

4. On carriage side again (left), EE wrap on (2" of stitches).
 5. Hang comb or weights on those stitches and knit across.
- Make sure all stitches knit.

Stitch Gauge	7	7 ½	8	8 ½	
Add Stitches LEFT ←	14	15	16	17	



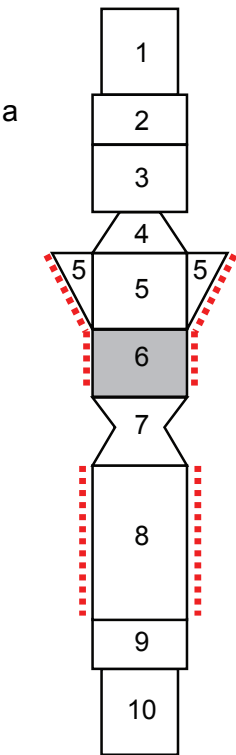
7. Program machine for Fair Isle Pattern, **excluding the three stitches at each side.**
8. Set carriage to Fair Isle tension.
9. Knitting in pattern, using your 3-prong tool, DECREASE 1 stitch each side every other row until you have decreased all the stitches you added.



Stitches remaining after decreases					
Stitch Gauge	7	7 ½	8	8 ½	
Stitches	27	28	29	31sts	

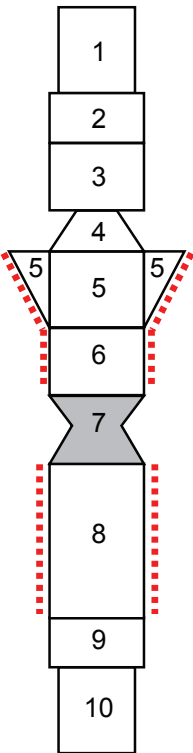


Section 6: Sole



Continue knitting in pattern until row counter shows total of (6.5" of rows) for sections 5 and 6

Row Gauge	Rows Section 5 and 6
9	58
9 ½	62
10	66
10 ½	68



Section 7: Toe

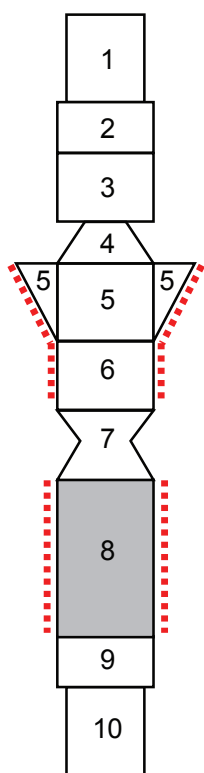
- 1. Change to Stockinette stitch and stockinette tension
- 2. Short row for (1.75" of rows)
- 3. Reverse Short row until all stitches are back in work.

Row Gauge	Rows to Knit
9	16
9 ½	16
10	18
10 ½	18



The Ultimate Machine Knit Socks

Parsley Single Bed | Standard-Gauge



Section 8: Instep

Knit in Stockinette for (6. 5" of rows)

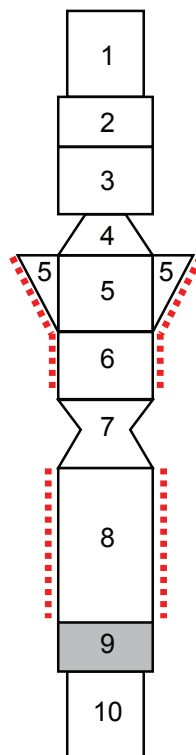
AT THE SAME TIME,

SEW-AS-YOU-GO

THE SIDE EDGES OF

SECTIONS 5 AND 6

Row Gauge	Rows to Knit
9	58
9 ½	62
10	66
10 ½	68

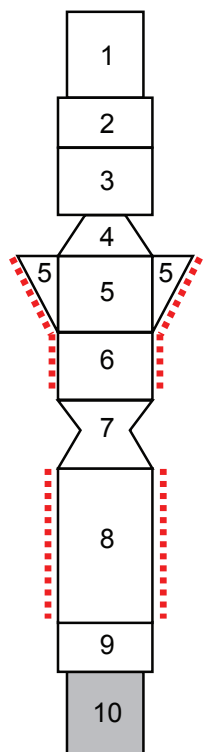


Section 9: Ankle

Knit to match section 2

(1" of rows)

Row gauge	Rows to Knit
9	9
9 ½	10
10	10
10 ½	11

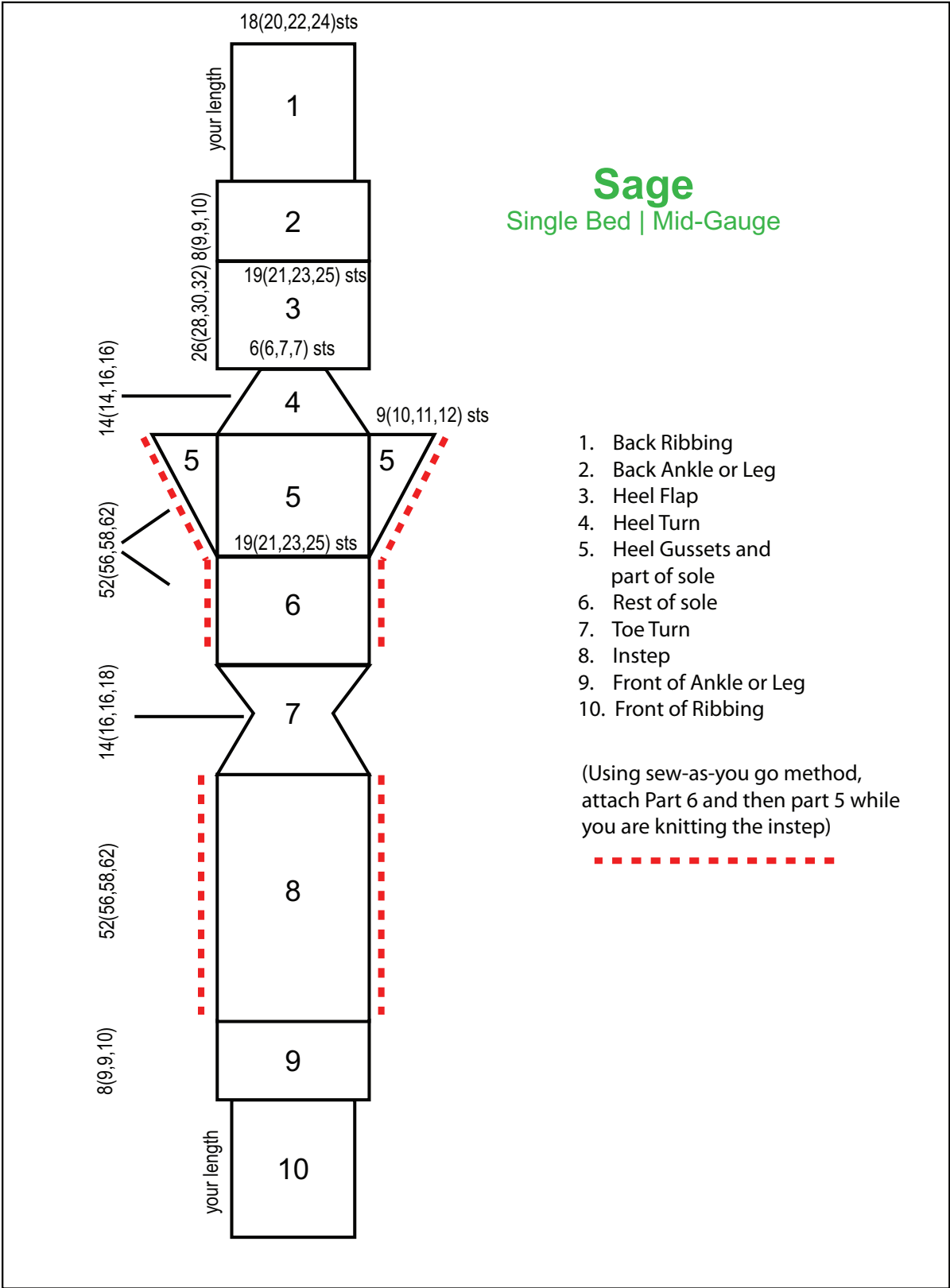


Section 10: Ribbing

1. Transfer to ribbing arrangement and knit to match section 1.
2. Knit the last row at maximum tension in ribbing and main yarn.
3. Knit several rows of waste yarn and remove from the machine.
4. Work a slip-stitch finish on last row and remove waste yarn.

Almost done! Follow the finishing instructions.





The Ultimate Machine Knit Socks

Sage Single Bed | Mid-Gauge

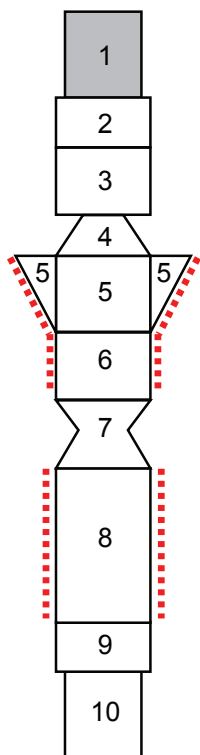
SAGE
WOMAN'S MEDIUM SOCK

With Plain Foot and Plain Heel

GAUGES:

5 (5 ½, 6, 6 ½) stitches per inch

8 (8 1/2 , 9, 9 1/2) rows per inch



The measurements for this medium size are in **GREEN**. See instructions for customizing your sock.

Section 1: Ribbing

EE wrap cast on double bed for 1x1 ribbing. Hang comb and weights. Knit desired length of ribbing. *(This is ½ of the ribbing. Section 10 is the other half. If you do not like a seam on both sides of the ribbing, begin with waste yarn and a row of ravel cord and knit sections 2 through 9. Then go back and knit the ribbing by hand or machine.)*

Cast on (3.6" of stitches)

Stitch Gauge	5	5 ½	6	6 ½	
Cast on	18	20	22	24	

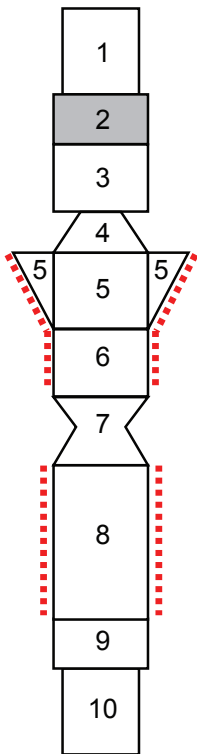


When knitting your first sock, we HIGHLY recommend knitting it entirely in stockinette. This will allow you to get a feel for the construction techniques, without having to worry about stitch patterning.



Ultimate Machine Knit Socks

Sage Single Bed | Mid-Gauge

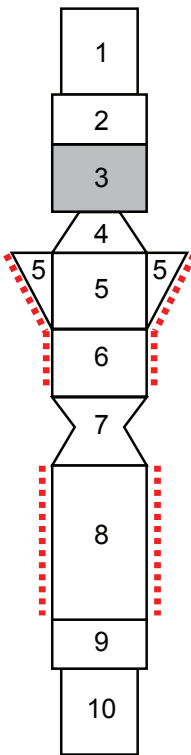


Section 2: Ankle

Knit in Stockinette

(1" of rows)

Row Gauge	Rows to Knit
8	8
8 ½	9
9	9
9 ½	10



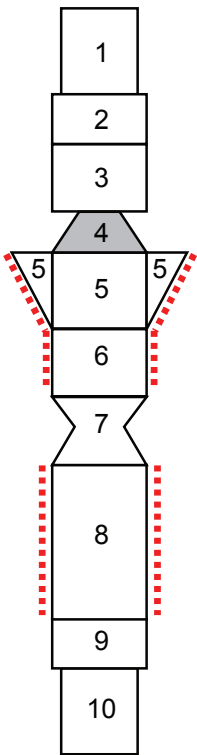
Section 3: Heel Flap

- You need to have an uneven number – increase 1 stitch if necessary

Stitch Gauge	5	5 ½	6	6 ½	
Stitches in work	19	21	23	25	

- Program machine for heel stitch.
- Knit for (2.25" of rows), changing colors every 2 rows

Row Gauge	Rows to Knit
8	26
8 ½	28
9	30
9 ½	32



Section 4: Heel Turn

Knit in Stockinette, working INCREASING short rows (1.5" of rows)

NOTE: The number of rows needs to be rounded to an even number so your heel turn will be the same on both sides.

Row Gauge	Rows to Knit	Starting Hold Stitches
8	14	6
8 ½	14	6
9	16	7
9 ½	16	7



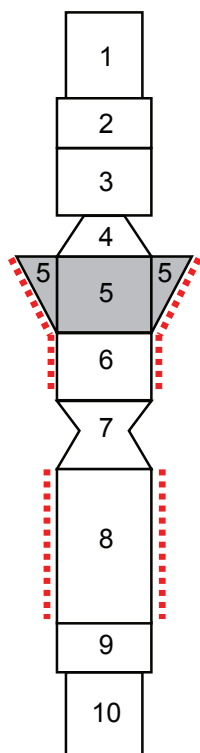
The Ultimate Machine Knit Socks

Sage Single Bed | Mid-Gauge

Section 5: Gussets and first part of sole

Reset Row counter to 000. Carriage on the right.

You will be knitting the foot length. The foot length for this size is (6.5" of rows)



1. On the carriage side (right), EE wrap cast on (2" of stitches)
2. Hang comb or weights on those stitches and knit 1 row.
3. Make sure all stitches knit.

Stitch Gauge	5	5 ½	6	6 ½	
Add Stitches RIGHT →	9	10	11	12	

4. On carriage side again (left), EE wrap on (2" of stitches)
5. Hang comb or weights on those stitches and knit across.

Stitch Gauge	5	5 ½	6	6 ½	
Add Stitches LEFT ←	9	10	11	12	

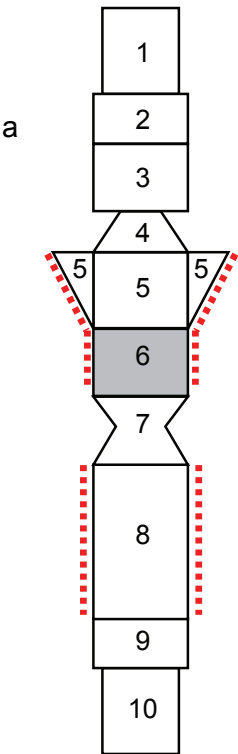
6. Make sure all stitches knit.
7. Using your 3 pronged tool, DECREASE 1 stitch, both sides, every other row until you have decreased all the stitches you added

Stitches remaining after decreases					
Stitch Gauge	5	5 ½	6	6 ½	
Stitches	19	21	23	25	



Ultimate Machine Knit Socks

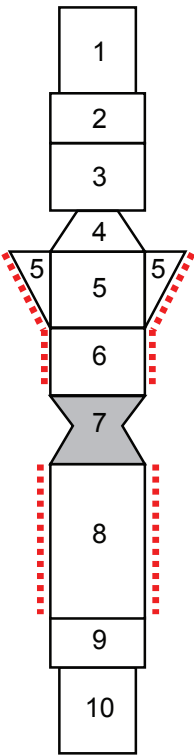
Sage Single Bed | Mid-Gauge



Section 6: The rest of the sole

Continue knitting in pattern until row counter shows total of (6.5" of rows) for sections 5 and 6

Row Gauge	Rows to Knit
8	52
8 ½	56
9	58
9 ½	62



Section 7: Toe

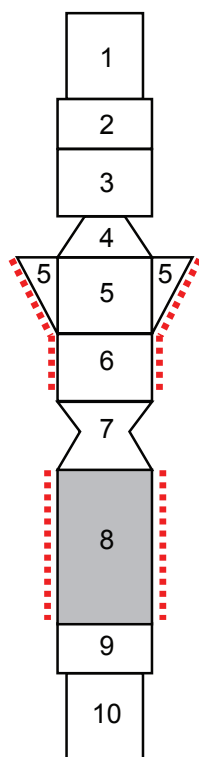
1. Change to Stockinette stitch and stockinette tension
2. Short row for (1.75" of rows)
3. Reverse Short row until all stitches are back in work.

Row Gauge	Rows to Knit
8	14
8 ½	16
9	16
9 ½	16



The Ultimate Machine Knit Socks

Sage Single Bed | Mid-Gauge



Section 8: Instep

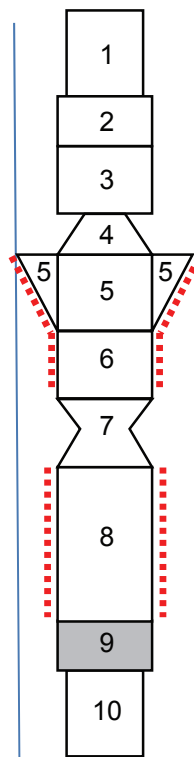
Knit in Stockinette for (6. 5" of rows)

AT THE SAME TIME,

SEW-AS-YOU-GO

THE SIDE EDGES
OF SECTIONS 5 AND 6

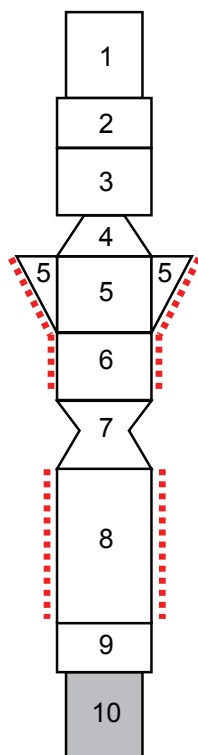
Row Gauge	Rows to Knit
8	52
8 ½	56
9	58
9 ½	62



Section 9: Ankle

Knit to match section 2
(1" of rows)

Row Gauge	Rows to Knit
8	8
8 ½	9
9	9
9 ½	10



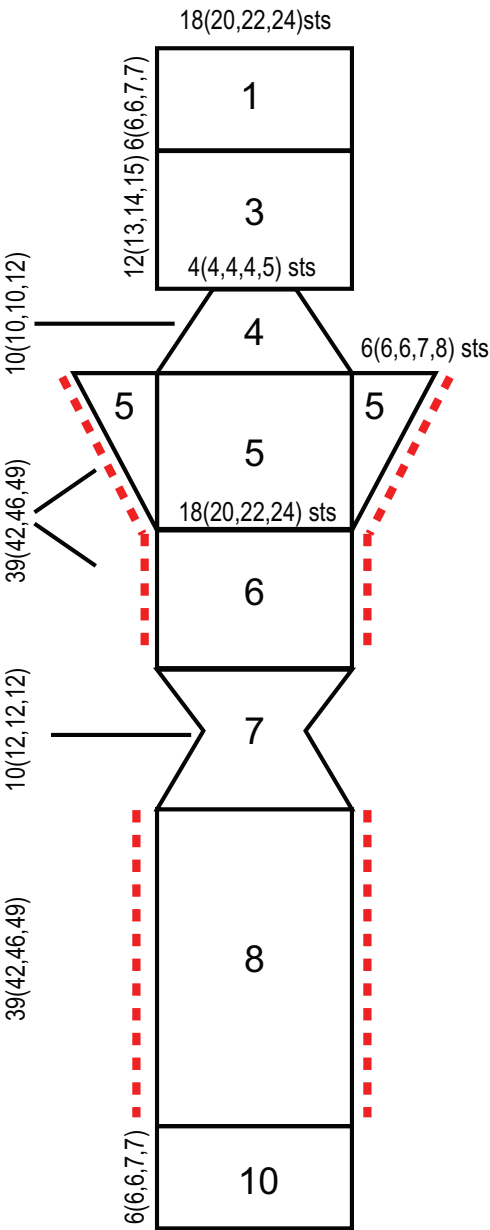
Section 10: Ribbing

1. Transfer to ribbing arrangement and knit to match section 1.
2. Knit the last row at maximum tension in ribbing and main yarn
3. Knit several rows of waste yarn and remove from the machine
4. Work a slip stitch finish (stretchy bind off)

Almost done! Follow the finishing instructions.



Rosemary
Single Bed | Bulky/Chunky Gauge



- 1. Back Ribbing
- 3. Heel Flap
- 4. Heel Turn
- 5. Heel Gussets and part of sole
- 6. Rest of sole
- 7. Toe Turn
- 8. Instep

10. Front of Ribbing

(Using sew-as-you go method, attach Part 6 and then part 5 while you are knitting the instep)

.....



The Ultimate Machine Knit Socks

Rosemary Single Bed | Bulky/ Chunky

ROSEMARY

WOMAN'S MEDIUM HOUSE SOCK

WITH Plain heel flap, reverse stockinette foot and heel turn with no ribbing.

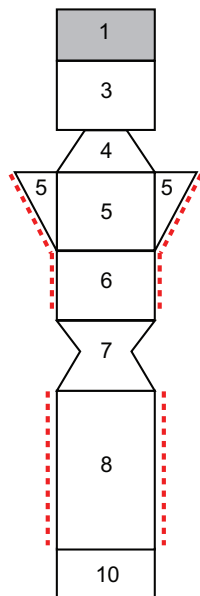
GAUGES:

3 $\frac{3}{4}$ (4, 4 $\frac{1}{2}$, 5) stitches per inch

6 (6 $\frac{1}{2}$, 7, 7 $\frac{1}{2}$) rows per inch

The measurements for this medium size are in **GREEN**. See instructions for customizing your sock.

Stitch Gauge	3 $\frac{3}{4}$	4	4 $\frac{1}{2}$	5 $\frac{1}{2}$	
Cast On	18	20	22	24	



Section 1: Back Hem (no ribbing)

Cast on with waste yarn and knit 10-20 rows. Knit 1 row of ravel cord.

1. Knit (1" of rows).
2. Make a turning row with Double E-wrap
3. Set carriage for a free pass and move it across to the opposite side (by the working yarn)
4. Replace yarn in feeder and knit 1" of rows

Section 2: No ANKLE

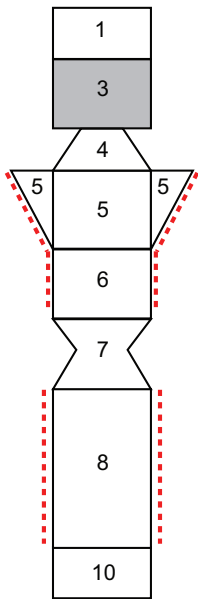


1" of rows	
Row Gauge	Rows to Knit
6	6
6 $\frac{1}{2}$	6
7	7
7 $\frac{1}{2}$	7



Section 3: Plain Heel Flap

- 1. Knit in Stockinette for (2" of rows)



For reverse stockinette foot:

- 1. Remove on waste yarn or garter bar.
- 2. Turn work over and rehang on the same needles.

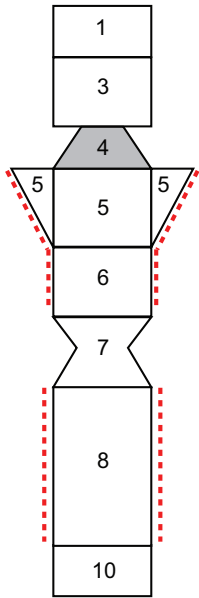
2" of rows	
Row Gauge	Rows to Knit
6	12
6 ½	13
7	14
7 ½	15



Section 4: Heel Turn

- 1. Knit in Stockinette, working INCREASING short rows (1.5" of rows)

NOTE: The number of rows needs to be rounded to an even number so your heel turn will be the same on both sides.



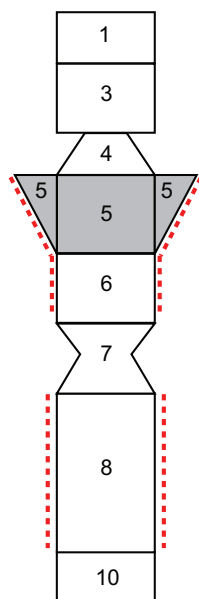
Row Gauge	Rows to Knit	Starting Hold Stitches
6	10	4
6 ½	10	4
7	10	4
7 ½	12	5



The Ultimate Machine Knit Socks

Rosemary Single Bed | Bulky/ Chunky

Section 5: Gussets and first part of sole



Reset Row counter to 000

1. On the carriage side (right), EE wrap cast on (2" of stitches)
2. Hang comb or weights on those stitches and knit 1 row.

3. Make sure all stitches knit.

4. On carriage side again (left), EE wrap on (2" of stitches)

5. Hang comb or weights on those stitches and knit across

6. Make sure all stitches knit.

7. Using your 3 pronged tool, DECREASE 1 stitch, both sides, every other row until you have decreased all the stitches you added

Stitch Gauge	3 ¾	4	4 ½	5 ½	
Add Stitches RIGHT →	6	6	7	8	

Stitch Gauge	3 ¾	4	4 ½	5 ½	
Add Stitches LEFT ←	6	6	7	8	

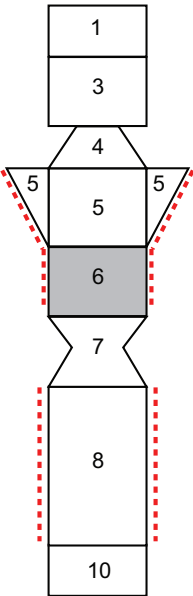


Stitches remaining after decreases					
Stitch Gauge	3 ¾	4	4 ½	5 ½	
Cast On	18	20	22	24	



Ultimate Machine Knit Socks

Rosemary Single Bed | Bulky-Gauge



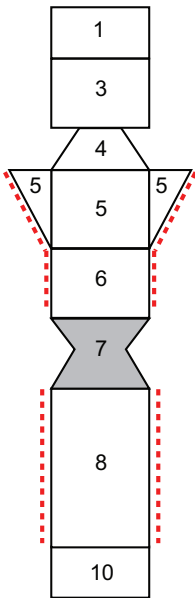
Section 6: The rest of the sole

Continue knitting in pattern until row counter shows a total of (6.5" of rows) for sections 5 and 6

Row Gauge	Rows to Knit
6	39
6 ½	42
7	46
7 ½	49

For reverse stockinette foot:

1. Remove on waste yarn or garter bar.
2. Turn work over and rehang on the same needles.



Section 7: Toe

1. Change to Stockinette stitch and stockinette tension
2. Short row for (1.75" of rows)
3. Reverse Short row until all stitches are back in work.

Row Gauge	Rows to Knit
6	10
6 ½	12
7	12
7 ½	12



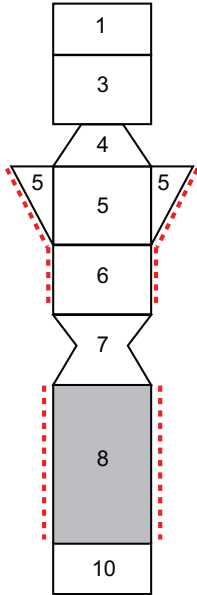
The Ultimate Machine Knit Socks

Rosemary Single Bed | Bulky/ Chunky

Section 8: Instep

1. Knit in Stockinette for (6. 5" of rows)

AT THE SAME TIME, SEW-AS-YOU-GO
THE SIDE EDGES OF SECTIONS 5 AND 6

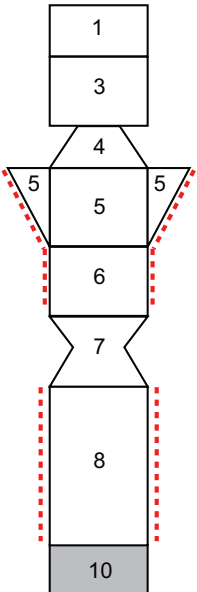


Row Gauge	Rows to Knit
6	39
6 ½	42
7	46
7 ½	49

Section 9: No ANKLE

Section 10: Front Hem

1. Make a hem and turning row as for the back hem
2. Remove on waste yarn



1" of rows	
Row Gauge	Rows to Knit
6	6
6 ½	6
7	7
7 ½	7

Almost done!

Follow the finishing instructions.

Hem finishing:

Stitch hems to the inside. If desired, you can insert elastic into the hems and then close the side edges. Or close the inside seam of hems, thread an I-cord or elastic shoe string or ribbon through the hems, and tie in a bow at the side.

NOTE: The foot for this version can also be knitted in Fairisle as in the pattern for the standard gauge machine. You can also add no-slip paint to the sole as desired.



Ultimate Machine Knit Socks

Single Bed Alternate Version (any Gauge)

Single Bed Alternate Version (any gauge)

It is recommended that you knit Version 1 first. This will help you understand the basic construction before you use this alternate method. This method enables you to knit the sock without needing to sew the tops of the gussets to the side edges of the flaps.

From Lucy

I figured this one out because I had forgotten to knit the heel turn (section 4). I had knitted as far as the heel flap and right away e-wrapped on for the gussets instead of knitting the heel turn first.

Looking at the silly thing I suddenly thought why not turn it around, hang the first row of section 5, knit the heel turn, then the heel flap as in so doing I could sew-as-you-go the e-wraps from the gussets and not have to sew them up later.



The Ultimate Machine Knit Socks

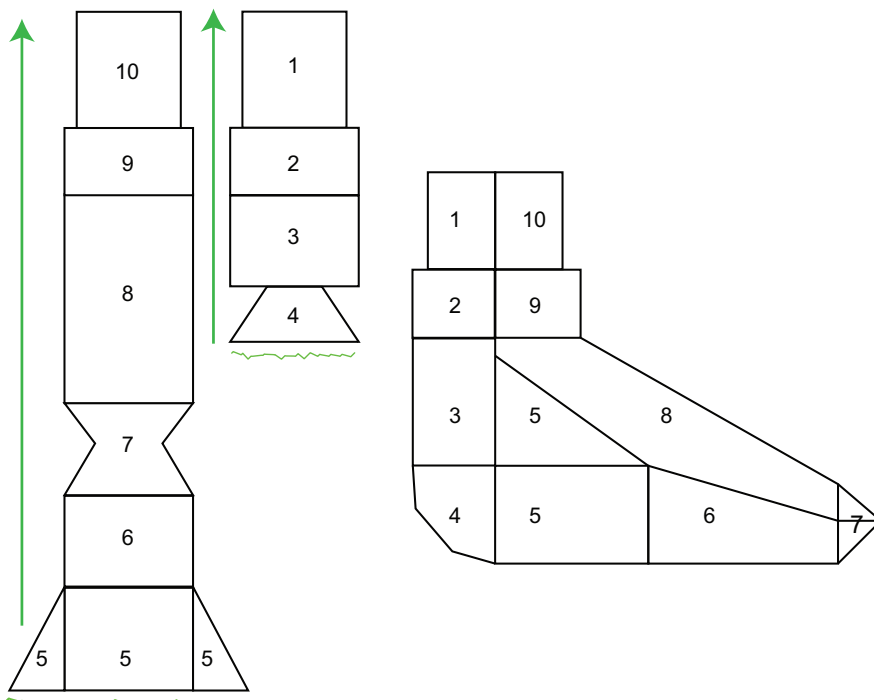
Single Bed Alternate Version

Order of Knitting:

Sections 5, 6, 7, 8, 9, 10

THEN

Sections 4, 3, 2, 1.



1. Cast on with waste yarn at section 5, including the gusset sections that are normally e-wrapped on. **Use separate sections of waste yarn for each gusset and the sole part.** Top of gussets now are regular open stitches instead of e-wraps.
2. Knit sections 5, 6, 7, 8 and end with waste yarn.
3. Hang the first row (sole part only) of section 5 and knit section 4 (heel turn) the same way you would knit the first half of the toe, i.e., a decreasing short row (set machine to hold, bring needle closest to carriage out to hold, knit across and repeat until you are holding one third of the stitches on each side.). Put held stitches opposite the carriage into work. Knit across. Put the held stitches at the other side back into work.
4. Now knit the heel flap and **AT THE SAME TIME** use sew-as-you-go to attach the open stitches from the gusset cast-on to the sides of the heel flap.
5. Follow the diagram: the knitting order is 5, 6, 7, 8, 9, 10, then 4, 3, 2, 1.

Don't forget to hang a bobby pin over the yarn coming from the end stitch at the beginning of each row when knitting sections 5 and 6. You will have a bobby pin in the loop on each side every other row. When you knit section 8, pick up the bobby pin and hang the loop it is in onto the end needle opposite the carriage. It truly makes these socks easier and guarantees no mistakes for sew-as-you-go.

From Lucy

*I also found I wanted the heel flap about half an inch longer so knitted 6 additional rows of flap and then took it off on waste yarn.
I added a few rows at the front ankle to match the extra heel flap rows when I knit the leg and ribbing.*



Ultimate Machine Knit Socks

Double Bed Circular Socks

Double Bed (Ribber) Version

This method incorporated the SWAN sock heel flap, heel turn and gusset construction.



Sue's Tip

PLEASE knit at least 1 single bed sock before trying this version.

Use measurements from single bed sock.

MANDATORY SWATCHES:

- Stockinette on the single bed.
- Circular/tubular on both beds.

You will have to make tension swatches to determine what settings to use to make front and back beds match each other.

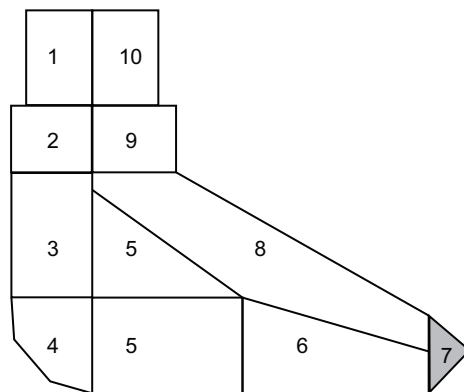
Back Bed – Main Bed

Front Bed - Ribber

Section 7: Toe Shaping

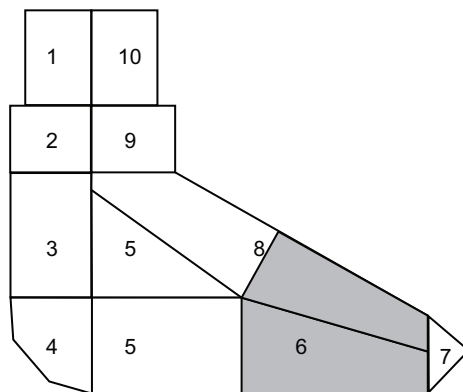
You will start knitting this version at section 7 on the diagram.

1. Cast on foot width on back bed with waste yarn and knit 20 rows.
2. Knit 1 row of ravel cord
3. Knit 1 row of main yarn, then shortrow for toe shaping.



Sections 6-8: Instep

4. On the front bed, put the same number of needles into work.
5. Hang the stitches from first main yarn row after ravel cord row onto the front bed.
6. Knit in a circular fashion on both beds (sections 6 and 8) until you reach the beginning of section 5.

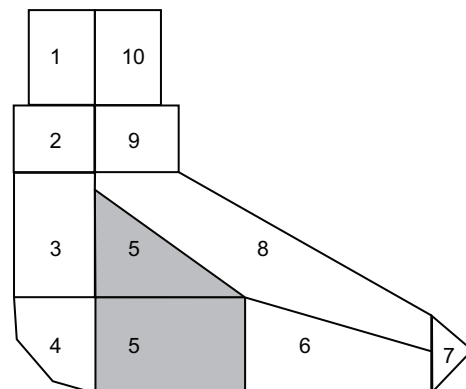


The Ultimate Machine Knit Socks

Double Bed | Circular Socks

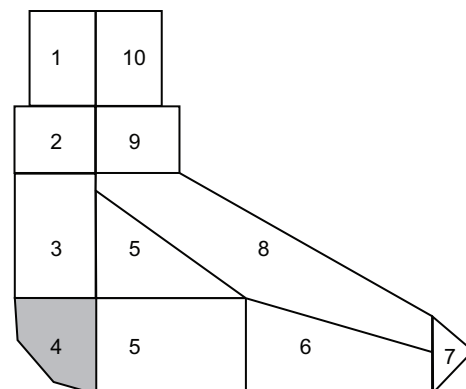
Sections 5-8: Gusset

7. **Increase 1 stitch each side on back bed by transferring 2 stitches at each edge and picking up a purl loop to fill the vacated stitch.
8. Knit 2 rounds.
9. Transfer the outermost stitch at each edge of back bed to the front bed, putting the stitch adjacent to stitches in work.
10. Increase 1 stitch each side on back bed as before.
11. Knit 2 rounds.**
12. Repeat from ** to ** until you have increased the required number of stitches for the gussets. (Section 5) Half of the increased stitches will be on the back bed and half will be on the front bed.
13. Remove all stitches onto waste yarn.



Section 4: Heel

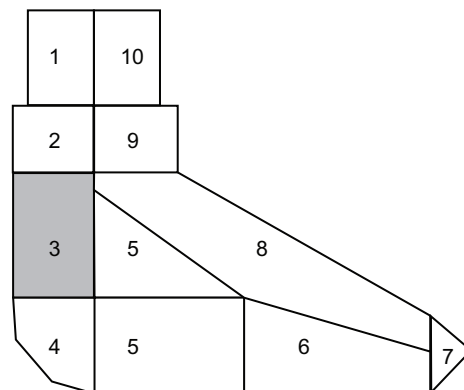
14. Drop front bed, change to single bed feeder.
15. Rehang on the back bed the stitches for sole (not the gusset or instep stitches)
16. Program for heel stitch pattern and do a decreasing short row for the heel turn. (Section 4).
17. Bring all back bed stitches back into work.



Section 3: Heel Flap

18. Continuing in heel pattern stitch, knit heel flap

It is possible to utilize the sew-as-you-go method and attach the stitches of the gussets to the side edges of the heel flap. It's actually better and easier to sew them later by hand.

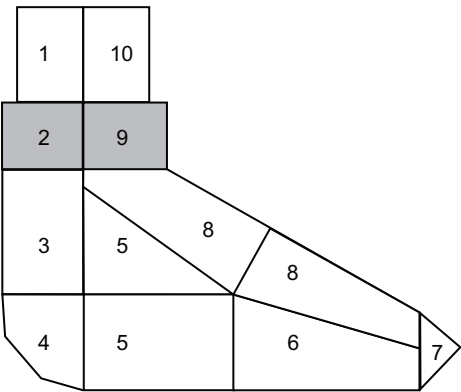


Ultimate Machine Knit Socks

Double Bed Circular Socks

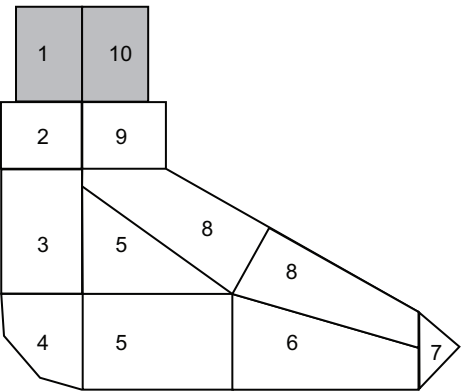
Sections 2-9: Ankle

19. Rehang the instep stitches on the front bed and knit "U" knitting on both beds for ankle. Do not knit in the round; it will be easier to knit the ribbing if you leave a seam in the ankle section.



Sections 1-10: Ribbing

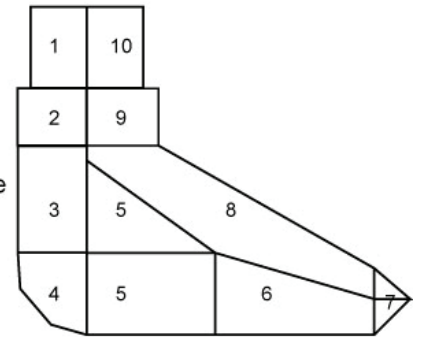
20. Remove stitches from front bed onto waste yarn and hang them adjacent to the stitches on the back bed
21. Transfer to ribbing formation and knit ribbing.



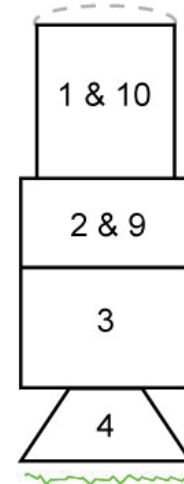
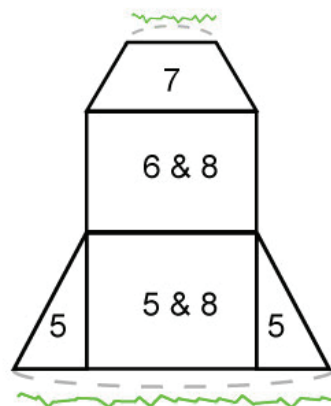
Quick Gusset Version Double Bed (Ribber)

This method gives you the same instep room as the traditional gusset while giving less interference with self-stripping patterns

1. Back Ribbing
2. Back Ankle or Leg
3. Heel Flap
4. Heel Turn
5. Heel Gussets and Part of Sole
6. Remainder of Sole
7. Toe Turn
8. Instep
9. Front Ankle or Leg
10. Front Ribbing



1. Cast on with waste yarn and ravel cord at section 5, including gussets, plus section 8 (instep). Divide the stitches equally between the beds so that half the gusset stitches are on each bed at each side of the sock.
2. To decrease the gussets use your 2 prong tool and decrease 1 stitch each side, each bed, then knit 4 rounds (8 rows counted).
3. Continue decreasing until all gusset stitches are gone.
4. Knit sections 6 and 8 circularly (sole and instep)
5. To knit the toes, decrease 1 stitch each side, each bed
6. Knit 2 rounds
7. Repeat until you've finished the toe.
8. Remove on waste yarn and graft the toe to finish
9. Hang the stitches from the first row of the instep (section 8) onto the front bed.
10. Knit circularly the front and back ankle or leg.



Sue's Tip

You can work the decreases at the edge of the sole and instep stitches for a different look.

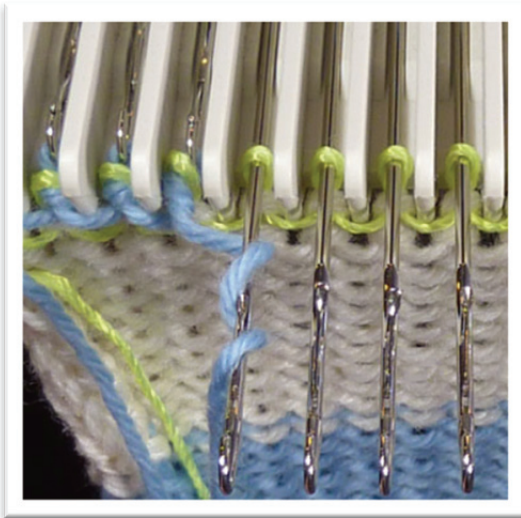
For instance, if you have 16 gusset stitches (8 on each side, each bed), the first decrease would be to move 7 stitches in, each side, each bed. Knit 4 rounds (8 rows total), then move 6 stitches in, etc.



Techniques used the patterns:

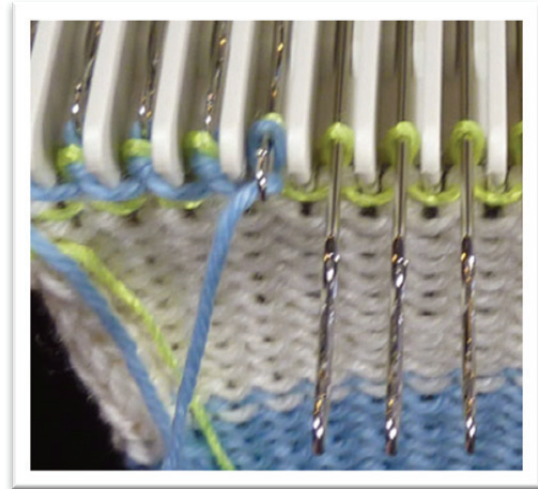
Double E-wrap (Secured e-wrap)

1. Pull empty needle forward as far as it will go.
2. E-wrap the yarn in a counter clockwise direction once around the needle behind the latch, the again in the needle hook.
3. Pull the needle back into working position, knitting the yarn in the hook through the wrap of yarn that is behind the latch.



Sue's Tip

Add a cast on rag (or waste knitting) and a row of ravel cord before working the Double E-wrap. Use the rag to hang a weight. This makes knitting these cast on stitches easier.



The Ultimate Machine Knit Socks

Techniques

Increasing short rows for heels (with automatic wrap)

This is an example. Start your increases with the number of needles indicated in your pattern under the column "Starting Hold Stitches."

Row 1: (carriage on the right) Opposite the carriage, put 9 needles into hold
AND
knit across.

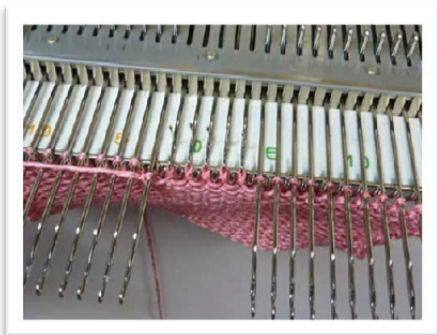
Row 2: (carriage on the left) On the carriage side, put one MORE needle in hold (10 needles now in hold)
AND
opposite the carriage, put 9 new needles in hold
AND
knit across.

Row 3: (carriage on the right) On the carriage side, put one MORE needle in hold (10 needles now in hold on each side (total of 20 needles in hold)
AND
on the side opposite the carriage, put the 2 needles closest to 0 into work. (You putting stitches into working position in the middle section of the knitting, keeping the needles on the far right and left in hold)
AND
knit across.
You will have 10 needles in hold on the right and 8 needles in hold on the left.

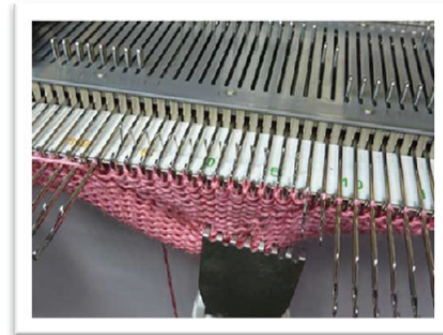
You can see that this actually increases the number of needles knitting by one stitch each side, every other row. Thus it is an increasing short row process

Row 4: (carriage on the left)
On the carriage side, put one needle back in hold position (for the automatic wrap)
AND
On the side opposite the carriage, put the 2 needles closest to 0 into work
AND
Knit across.
You will have 9 needles in hold on the left and 8 needles in hold on the right.

Continue in this manner, on each row bringing one more needle into hold on the carriage side and 2 into work on side opposite carriage, until all needles are back in work.



Needle L3 in hold to wrap Needles R4 In hold
Needles L2 – R3 in work



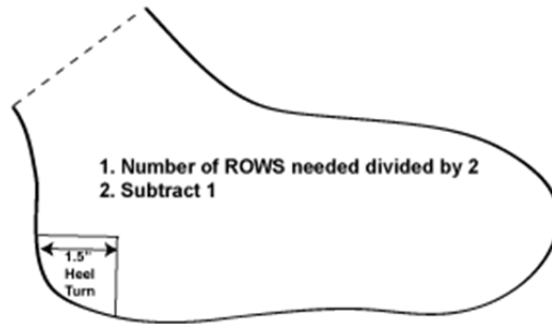
Automatic Wrap on carriage side
2 needles in working position opposite carriage



The Ultimate Machine Knit Socks

Techniques

If you are drafting your own sock (i.e., using your own numbers), you will need to determine the **number of stitches** to hold for the first row of your increasing heel turn.



EXAMPLE:

For example, if you are knitting 20 rows for the 1.5" heel turn, calculate the number of needles needed and follow the increasing short rows technique.

1. 20 divided by 2, minus 1. ($20/2=10-1=9$) put **9** needles out to hold opposite the carriage.
2. Knit across, put one more needle (next to the ones in hold) out to hold. Now you have 10 needles in hold on carriage side.
3. This gives you an automatic wrap to prevent holes.
4. Again, put 9 needles out to hold opposite the carriage.
5. Knit across, put one more needle out to hold on carriage side as an automatic wrap. Now you have 20 needles in hold, 10 at each side of your knitting.
6. On side opposite carriage, put the 2 needles closest to zero into work. Knit across.
7. Put one needle back into hold. Now you have 19 needles in hold, 9 at carriage side and 10 opposite carriage.

If desired, you can use an additional manual wrap on each needle at the carriage side before knitting across. I would only recommend this for the standard gauge machine.

Continue in this manner, on each row bringing one more needle into hold on carriage side and 2 into work on side opposite carriage, until all needles are back in work

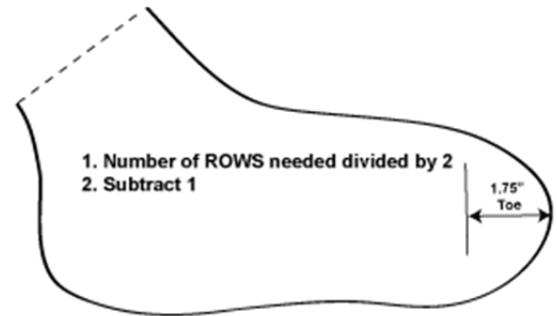


Toe Shaping

Short row and reverse short row for toes.

This method is an automatic wrap and makes a nice short rowed toe without holes.

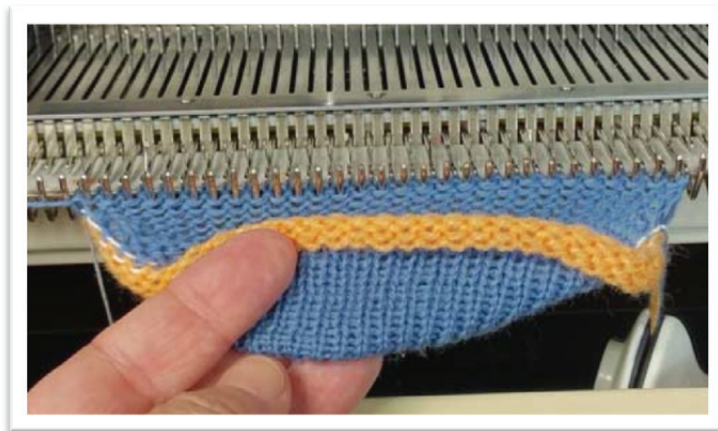
1. Fix machine so it will not knit the needles placed in hold.
2. On carriage side, pull edge most needle into hold and knit across.
3. Repeat until desired number of needles are in hold.
(1.75" times your row gauge divided by 2 minus 1)
4. You will short row for 1.75 inches of rows.



REVERSE SHORT ROW

5. On side opposite carriage bring 1 needle into work and knit across.
6. Continue until all needles are back in work.

Use claw weight and adjust frequently.



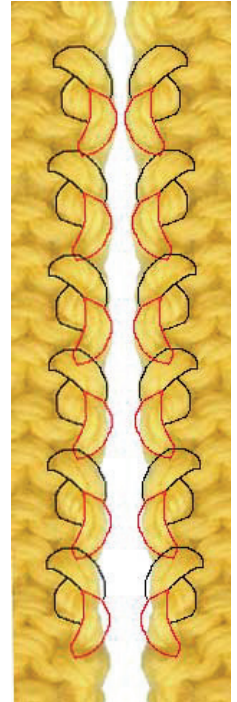
The Ultimate Machine Knit Socks

Techniques

Sew-as-you-go seaming method

The sole of your sock is knitted, the toe is done and you are ready to knit the instep or top of your foot. Before you knit each row, on the side opposite the carriage, you hang the loop from the outside edge of the sole onto the outside needle of the instep row you are about to knit. Knit across and repeat until the entire instep is knitted and the sole is attached at each side. When work is off the machine, give a hard crosswise pull all along the seam to even things out and to achieve the look of a mini cable.

If you look at the purl side of a piece of knitting and look at the edge, you will see that at each row there is what looks like a knot. Between each knot there is what looks like a longer loop of yarn. The loop is actually half of a stitch and the loop is what you hang. Be sure there is a knot on each side of the loop you are hanging and slip your eyelet tool into the loop to hang it on the needle. Be sure you see the knot on each side to avoid hanging the same loop twice.



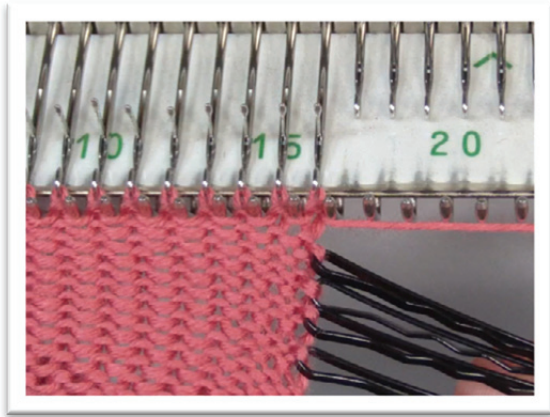
Sue's Tip

If you don't like the finished look of "sew-as-you-go", hand stitch the side seams with a Bickford Seam. It's flat, comfortable and neater looking.



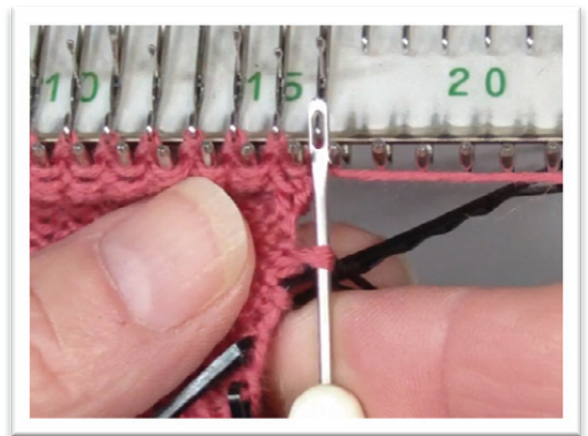
Bobby Pin Trick

When working sew-as-you-go seams, it can be difficult to identify the knots and loops. Some knitters find the bobby pin trick helpful. Mark each row as you are knitting the section you will hang. You can mark with yarn, plastic markers, bobby or hair pins, etc.



If you do this, hang your marker on the yarn leading to the carriage, right beside the last needle in work before you knit the next row. The marker is now in the loop you will hang so there is no question of what to hang. Take care to hold the yarn being fed into the carriage so you don't get a bigger loop because of the marker.

When you are ready to "sew-as-you-go," you can pick up the marker and hang the loop onto the outside needle.



The Ultimate Machine Knit Socks

Techniques

Stitch Patterns and Swatches

Tension Swatches: Swatching is NOT Optional!

To knit the sock as written, three tension swatches are needed.

SUGGESTION:

Find the stitch size you like for stockinette, then try one whole number tighter for the fairisle swatch. Use stockinette tension for the heel stitch swatch

1. **Stockinette stitch** and row gauge will be used to convert your measurements to your pattern.

2. **Fair Isle swatch** is needed to find which tension most nearly matches the row gauge of the stockinette swatch. Set your machine for 2 color knitting, use the stockinette tension minus 1 whole number.

4	●	●		
3	●	●		
2			●	●
1			●	●

3. **Heel Stitch Swatch** is needed to determine the number of rows for the heel flap. Use the Traditional or Eye of Partridge stitch pattern below.

Traditional Heel Flap Stitch

Slip setting, stockinette tension

Row 1: Knit entire row.

Row 2: Knit 1 stitch, slip 1 stitch.



4		●		●		●		●
3	●	●	●	●	●	●	●	●
2		●		●		●		●
1	●	●	●	●	●	●	●	●

Eye of Partridge heel stitch

Slip setting, stockinette tension

Row 1: Knit entire row.

Row 2: Knit 1 stitch, slip 1 stitch.

Row 3: Knit entire row.

Row 4: Slip 1 stitch, knit 1 stitch.



4	●		●		●	
3	●	●	●	●	●	●
2		●		●		●
1	●	●	●	●	●	●



How to make cast on rags

The original instructions for the Passap version of this sock use cast-on rags to form the gussets. This method could be used with any knitting machine.

These rags help you add a few stitches on the sides of your knitting, holding open stitches and providing knitting to hang weights on the newly formed stitches. Use waste yarn that will press well, like acrylic; you can *kill* the rags and the stitches won't ravel!

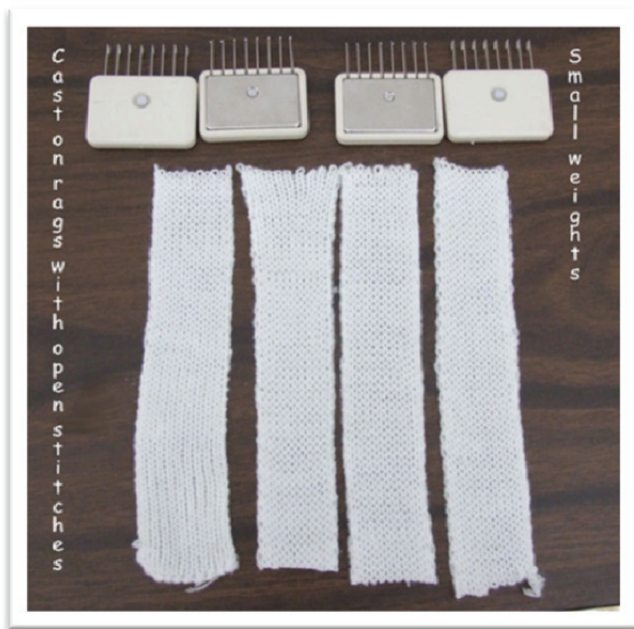
1. Cast on about 10 needles front bed in stockinette. Make large stitches.
2. Knit about 24"; no need to bind off.
3. Steam flat to set the stitches.
4. Cut into 4 equal parts.

Example: "Cast on 8 stitches on the left side of the bed"

With the carriage on the right, hang open loops of the cast on rag on 8 empty needles next to the knitting in work (you may want to hand knit a row of ravel cord for easier removal).

Knit across the work; you will have added 8 stitches to the row leaving open stitches.

When your knitting is complete, you will have the open stitches to finish as you like. This will eliminate the bulk of a closed cast-on for these few special stitches. You can re-hang them to continue knitting, or pick them up on a hand knitting needle as in the Passap sock instructions.



Sue's Tip

Not just for socks, cast-on rags are very useful whenever you need to add a few stitches on the edges of a row of knitting.

*Add underarm stitches when knitting top-down sweaters.
Add a gusset to a mitten or glove.*



The Ultimate Machine Knit Socks

Techniques

Taking Measurements

FOOT WIDTH

1. Measure closely around the foot at the SMALLEST point.
2. Reduce this number by 15% (multiply by 0.85).
3. Divide by 2.



FOOT LENGTH

1. Stand on a piece of paper, hold a ruler against heel at right angle to the paper and make a pencil mark on the paper.
2. Do the same at the point of the longest toe.
3. Measure the distance between the marks.
4. Subtract 1.5 inches for the heel turn.
5. Subtract 1.75 inches for toe turn.

Suggested measurements. Personal preference applies.

	Suggested Measurements	Your Measurements
Foot Width		
Foot Length		
Heel Turn	1.5"	
Toe Turn	1.75	
Heel Flap	2.00 – 2.25	
Beginning width of heel gusset	2" (standard gauge) 1.75" (mid-gauge) 1.5" (bulky)	
Length of gusset	Depends on your gauge	
Ankle length	1" - 1.5"	
Ankle Width	Same measurement as foot width (tighten tension for very slim ankles)	
Leg portion other than ribbing	Measure 2" above the ankle bone and subtract 10" (standard machine)	



The Ultimate Machine Knit Socks

Customizing your socks

Customizing your socks to fit YOU

To customize your sock, multiply your measurements by your stitch and row gauges and enter them in the blanks in the charts below.

Suggested measurements are shown below – edit as you feel necessary for your foot

	Stitch Gauge	Row Gauge	Stitches/Rows
Foot Width			sts
Foot Length			rows
Toe Length (1.75")			rows
Heel Turn (1.5")			rows
Width of heel Gussets (1.5 (Bulky) 1.75" (Mid-Gauge) 2" (Standard))			stitches
Heel Flap (2" or 2.25")			rows

Example: 7sts / 9 rows = 1"

		Stitch Gauge	Row Gauge	Stitches/Rows
Foot Width	8.5	X 7		60 sts
Foot Length	9		X 9	81 rows
Toe Length (1.75")	1.75		X 9	16 rows
Heel Turn (1.5")	1.5		X 9	14 rows
Width of heel Gussets (1.5 (Bulky) 1.75" (Mid-Gauge) 2" (Standard))	2	X 7		14 sts
Heel Flap (2" or 2.25")	2		X 9	18 rows

Substitute your custom stitch and row calculations for the perfect sock.



Ribbing Notes:

If you do not want to have a seam on both sides of the ribbing, cast on Section 1 and knit 10-20 rows with waste yarn and then knit 1 row of ravel cord and begin with Section 2. Follow instructions after Section 9 for completing the ribbing.

Option 1: Ribber

After completing Section 9, hang part 2 from the first row after the ravel cord, wrong side facing you, and adjacent to Section 9. Transfer to your preferred ribbing arrangement and knit ribbing (completing Sections 1 and 10)

Option 2: Hand knit

After completing Section 9, remove knitting from the machine on a hand knitting needle and pick up Section 2 stitches from the waste yarn. Hand knit the ribbing flat or in the round

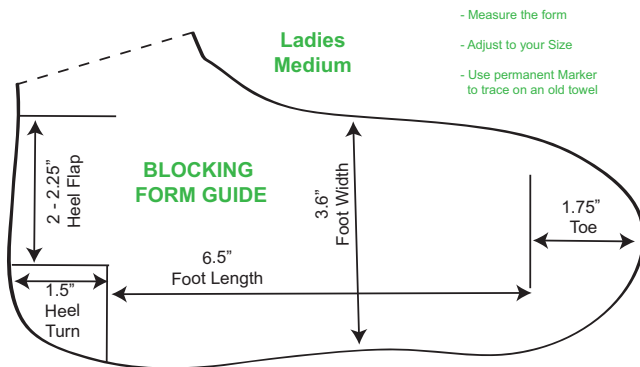


The Ultimate Machine Knit Socks

Finishing Instructions

Finishing Instructions

- 1.
2. Pull sideways on the sew-as-you-go seam. This pops the seam into a mini cable look.
3. Give a lengthwise and sideways pull to the entire sock to set the stitches.
4. Sew the side edge of heel flap to the cast-on gusset stitches. Be sure to sew the actual stitch from the first row of the gusset, not the EE wrap part.
5. Sew side seam and seams of ribbing.
6. Bury the yarn tails.
7. Block with gentle shot of steam. After the initial blocking, most yarns won't need to be blocked every time you launder them.

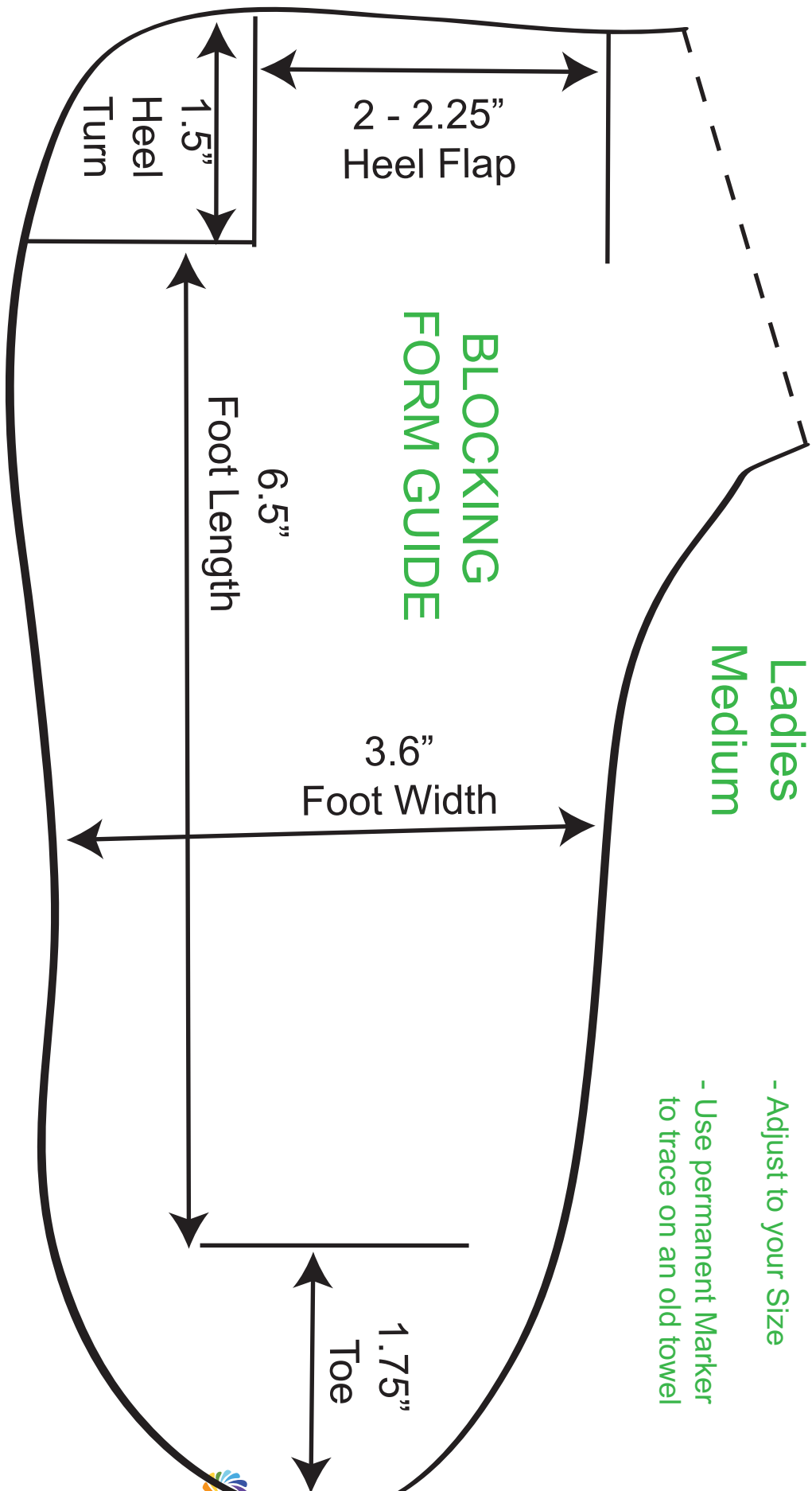


Full size form on next page.

Lucy's Tip

You can make a form for blocking. I think it's best to trace the form onto fabric with a permanent marker and use that as a guide as you steam. I use an old terrycloth towel. Or you can wash the sock and place it on the form to dry to shape. Let the yarn you use be your guide for blocking.





- Measure the form
- Adjust to your Size
- Use permanent Marker to trace on an old towel



The Ultimate Machine Knit Socks

Passap Duo 80 Tubular

SWAN Sock on the Passap DUO 80 Tubular

Sock is knit from the ribbing to the toe. All is tubular except the beginning rib; the ribbing is made tubular by use of Decker Combs.

Foot Width 3.62 x Stitch Gauge = stitches

Foot Length 9.75 times Row Gauge equals rows (RC)

Toe Length 1.5 times Row Gauge equals rows (RC)

Heel turn: 1.5 times Row Gauge equals rows

Width of heel gussets 2" times Stitch Gauge equals stitches

Divide this number by 2 = and this is the number of needles you add to each side of the foot width needles, both beds.

Heel Flap: 2" Heel stitch row gauge = rows.....row gauge for heel stitch is about $1\frac{1}{2}$ times the stockinette gauge.

1. Ribbing:

1. RC000 Cast on N/N, 1x1 rib, ORANGE STRIPPERS. You will be casting on the total number of stitches for the ribbing. Ribbing will be taken off on Decker Combs, transferred to the front bed and then from that point the sock will be knit tubular.
2. 1 row N/N Tension stitches .
3. 1 row CX/CX Tension .
4. Rib Tension for rows.
5. Unravel stitches (1/4 ribbing stitches) on the carriage side (unknit 1 row) .
6. Transfer all stitches to the back bed.
7. Take off on Decker Combs stitches, (1/4 # ribbing stitches) both ends. Turn Decker Combs like a page in a book and put stitches on front bed.

Now $\frac{1}{2}$ of stitches will be on back bed and $\frac{1}{2}$ on front bed with the seam opening center of front bed.



The Ultimate Machine Knit Socks

Passap Duo 80 Tubular

2. Ankle

1. RC000 – CX/CX – BLACK STRIPPERS Main Tension _____
 2. Knit _____ RC Note: I like about a 3" cuff.
-

3. Heel Flap

1. RC000 – back bed GX, front bed BX –BLACK STRIPPERS Tension _____
 2. Pusher set u
BX - XOXOXOXOX Knit 1 Row
N - XXXXXXXXXX Knit 1 Row
3. Repeat _____ RC _____

4. Heel Turn

1. RC000 – Back Bed GX, Front Bed BX – Tension _____ BLACK STRIPPERS
2. Put a pusher under every working needle. Allow 1.5" x row gauge _____ = _____ rows – divided by 2 = _____ number of pushers to put out of work on row 1 and row 2.
3. Auto wrap....put 1 less pusher in out of work position, knit 1 row, put 1 more pusher in out of work position when putting pushers back in work, put two pusher in WP, knit 1 row then put 1 pusher in out of work position RC _____.

5. Gusset

1. RC000 – Black Strippers CX/CX – Tension _____.
2. _____ stitches need to be added both ends, both beds.
3. Hang the cast-on rags on _____ needles each end, both beds.
4. Hand small weight on each rag (see note at end).
5. Hand knit the added stitches on the carriage side to bring the existing yarn out to the carriage.
6. Knit 2 rounds then start decreases for the gusset.
7. Decreases need to be done in 2" x row gauge; _____ = _____ rounds.
8. Decrease 1 stitch both beds, each end every _____ rounds.

****NOTE:** Make note of row count and rounds knitted to be used in Step 6. You should have left in work the original amount of stitches you had before adding the gusset stitches.

Rows

Rounds



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6. Foot

1. RC000 – CX/CX – BLACK STRIPPERS, Tension _____ MT.
2. Total foot length is _____" minus _____" for heel & toe equals _____" to be knit before toe decreases.

Equation....foot length _____" x _____" row gauge equals rounds _____ minus ** _____ rounds already knit in Step 5 equals _____ rounds to be knit. RC will read _____.

7. Toe:

1. RC000 – CX/CX – BLACK STRIPPERS Tension _____ MT.
2. Decrease 1 stitch full fashion both beds, each end _____ times EOR (RC4) then ER (RC2) _____ until 4 stitches remain.
3. Take off on double eye tool and draw up tight.



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8. Closing the Gusset

1. Pick up the gusset stitches on a US 1 double-pointed knitting needle. Snip the yarn and carefully unknit the cast-on rag; you can reuse them quite a few times.
2. Distribute the stitches on the knitting needle to match the heel flap. Pin the flap and gusset stitches together, and loosely crochet slip-stitch them together.
3. All that is left to do is sew up the ribbing, which will form a seam at the back, and work in the toe end.



Lucy's Tips

I've also done the gusset closing using Kitchener stitch. I think the Kitchener stitch is a better way to close the gusset.

Be sure to use a contrasting yarn for the cast on rags.





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NOTES

